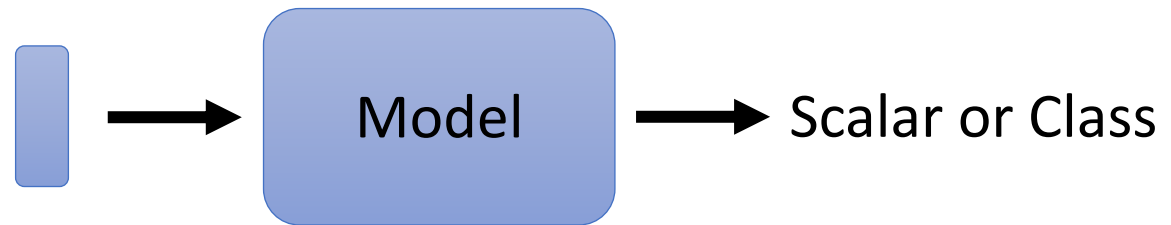


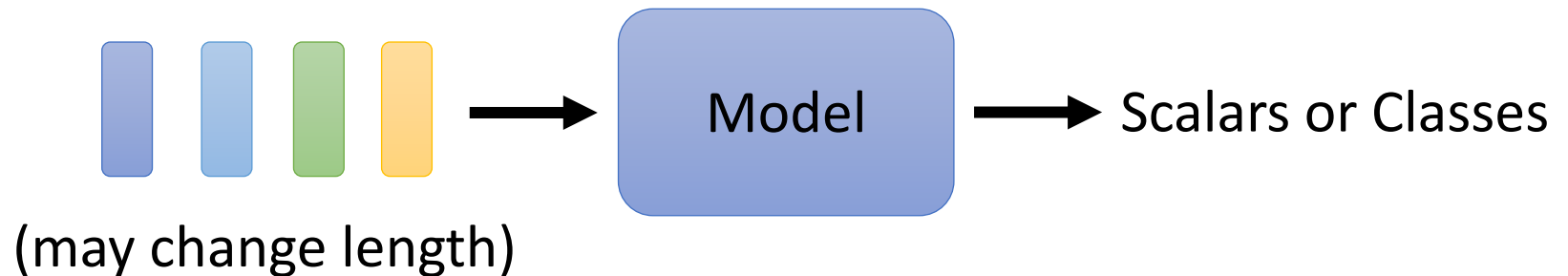
Self-attention

Sophisticated Input

- Input is a **vector**



- Input is a **set of vectors**



Vector Set as Input

this is a cat



One-hot Encoding

apple = [1 0 0 0 0]

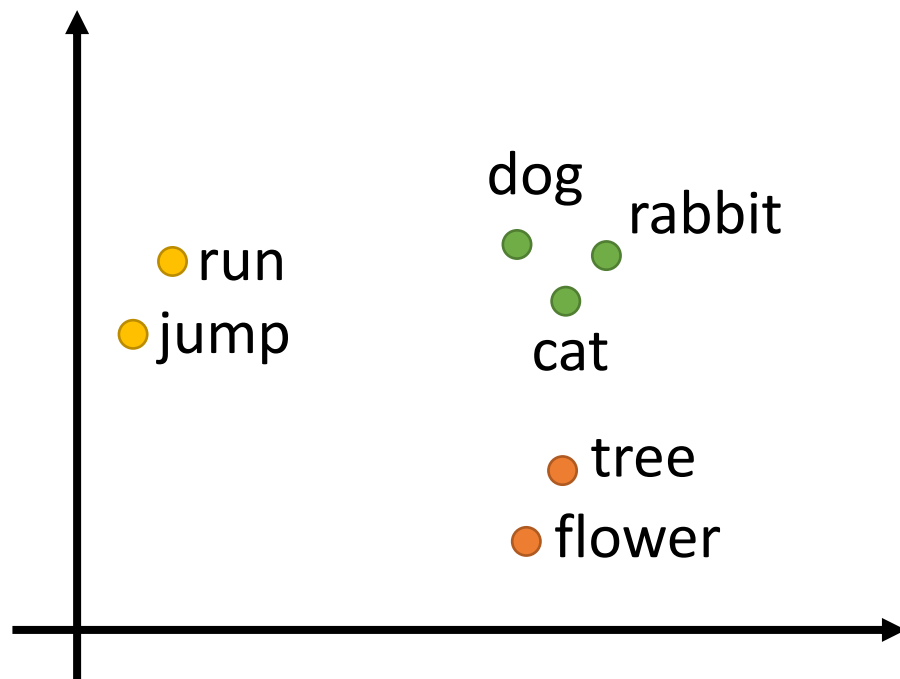
bag = [0 1 0 0 0]

cat = [0 0 1 0 0]

dog = [0 0 0 1 0]

elephant = [0 0 0 0 1]

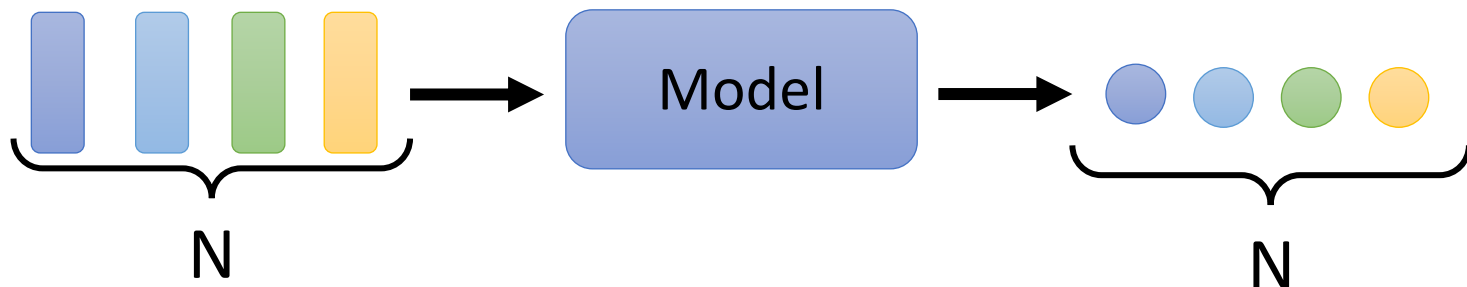
Word Embedding



To learn more: <https://youtu.be/X7PH3NuYW0Q> (in Mandarin)

What is the output?

- Each vector has a label.



Example Applications

I saw a saw

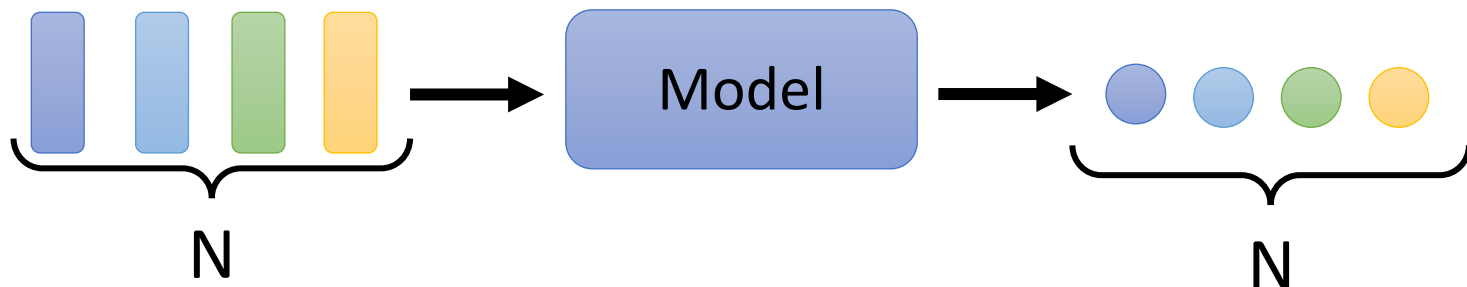
↓ ↓ ↓ ↓

N V DET N

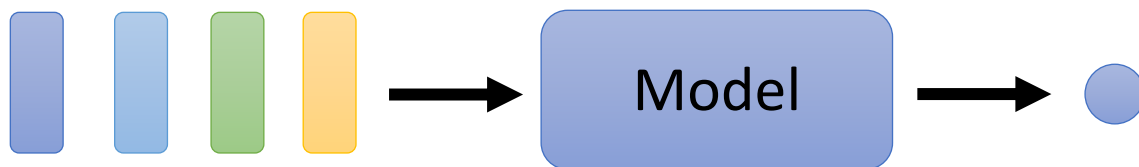
POS tagging

What is the output?

- Each vector has a label.



- The whole sequence has a label.



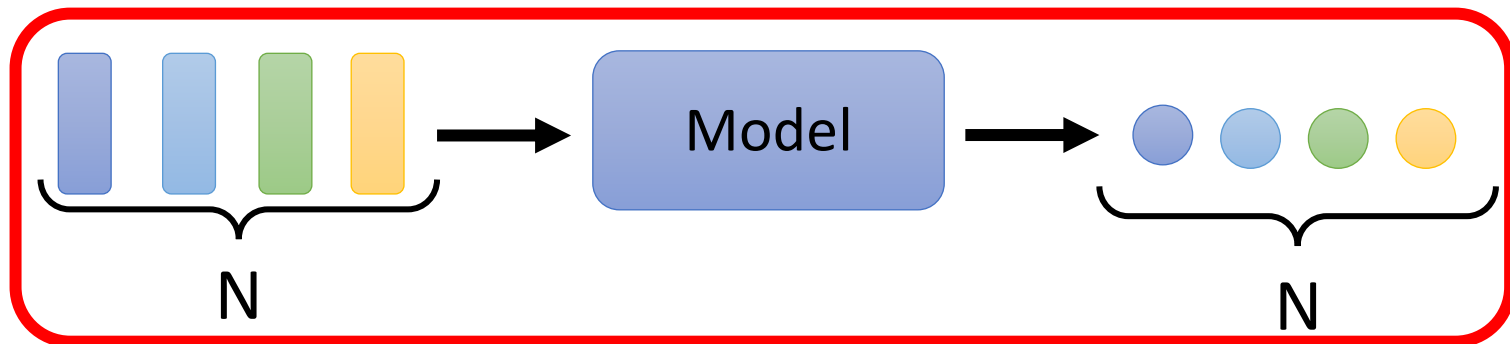
Example Applications

this is good
Sentiment
analysis
↓
positive

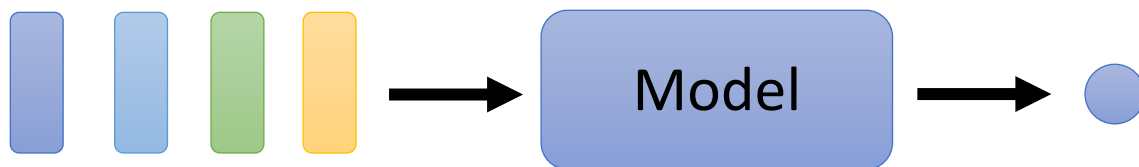
What is the output?

- Each vector has a label.

focus of this lecture

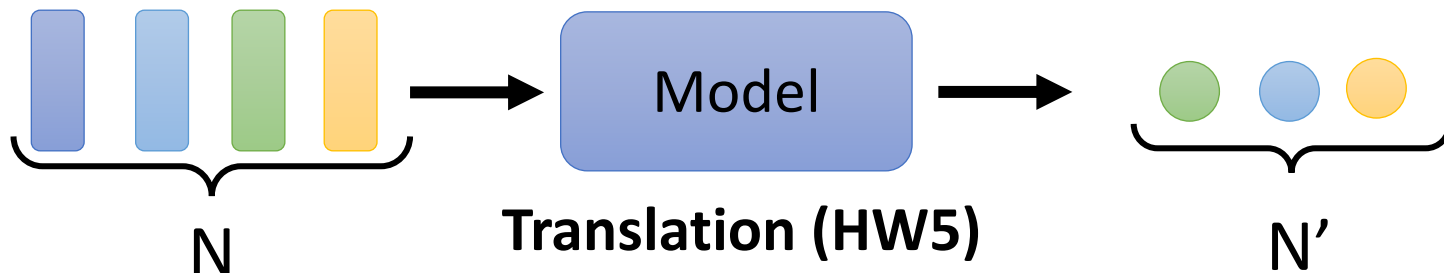


- The whole sequence has a label.



- Model decides the numbers of labels itself.

seq2seq



Sequence Labeling

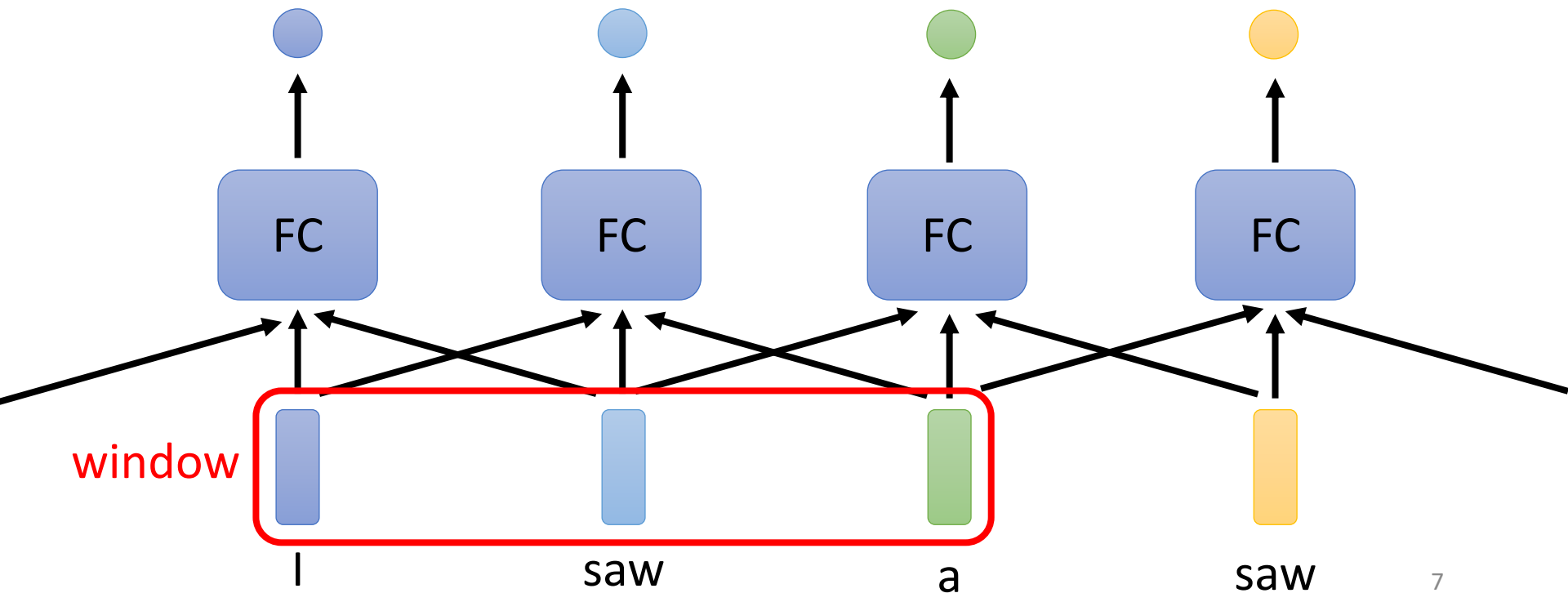
Is it possible to consider the context?

FC Fully-connected

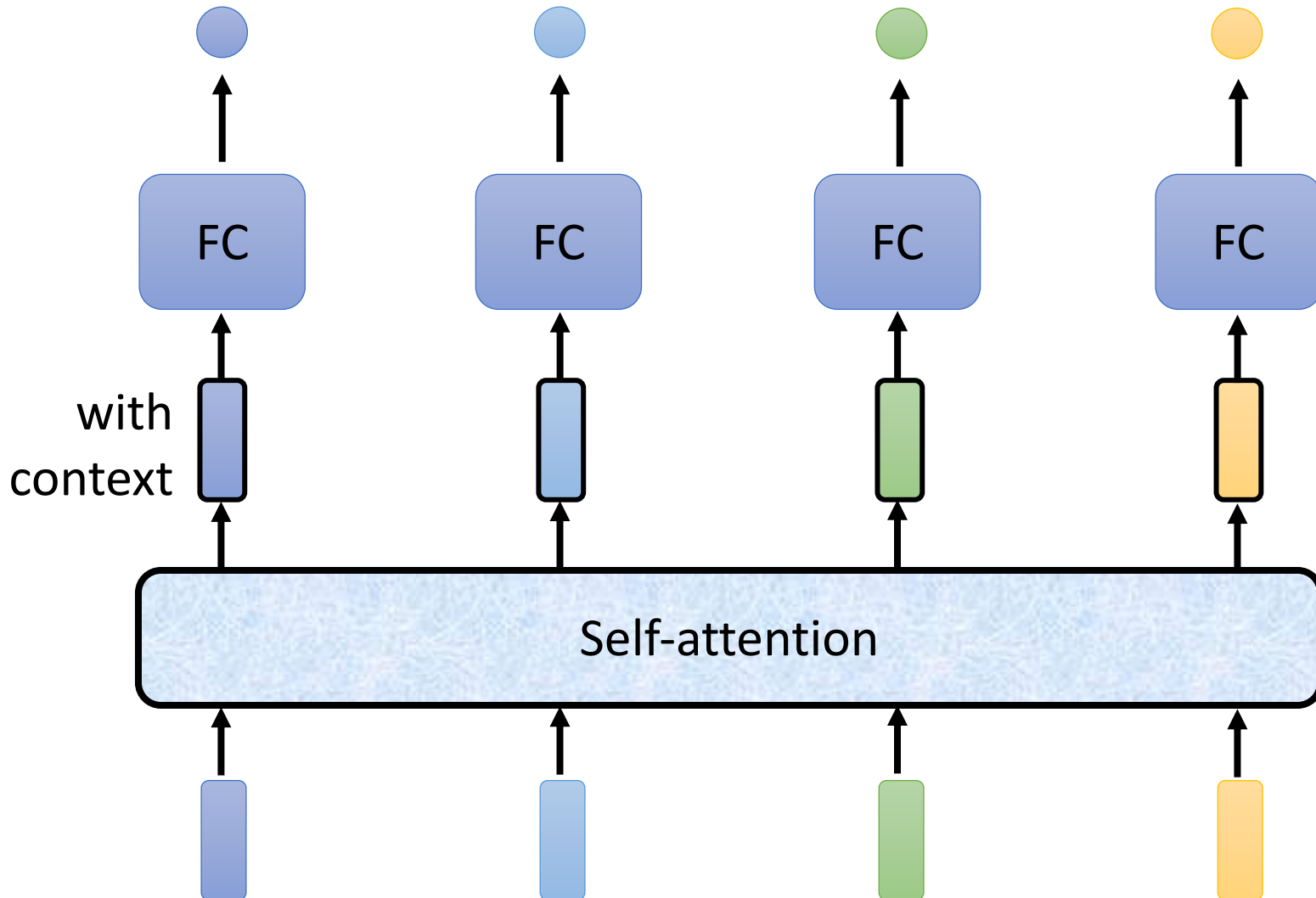
FC can consider the neighbor

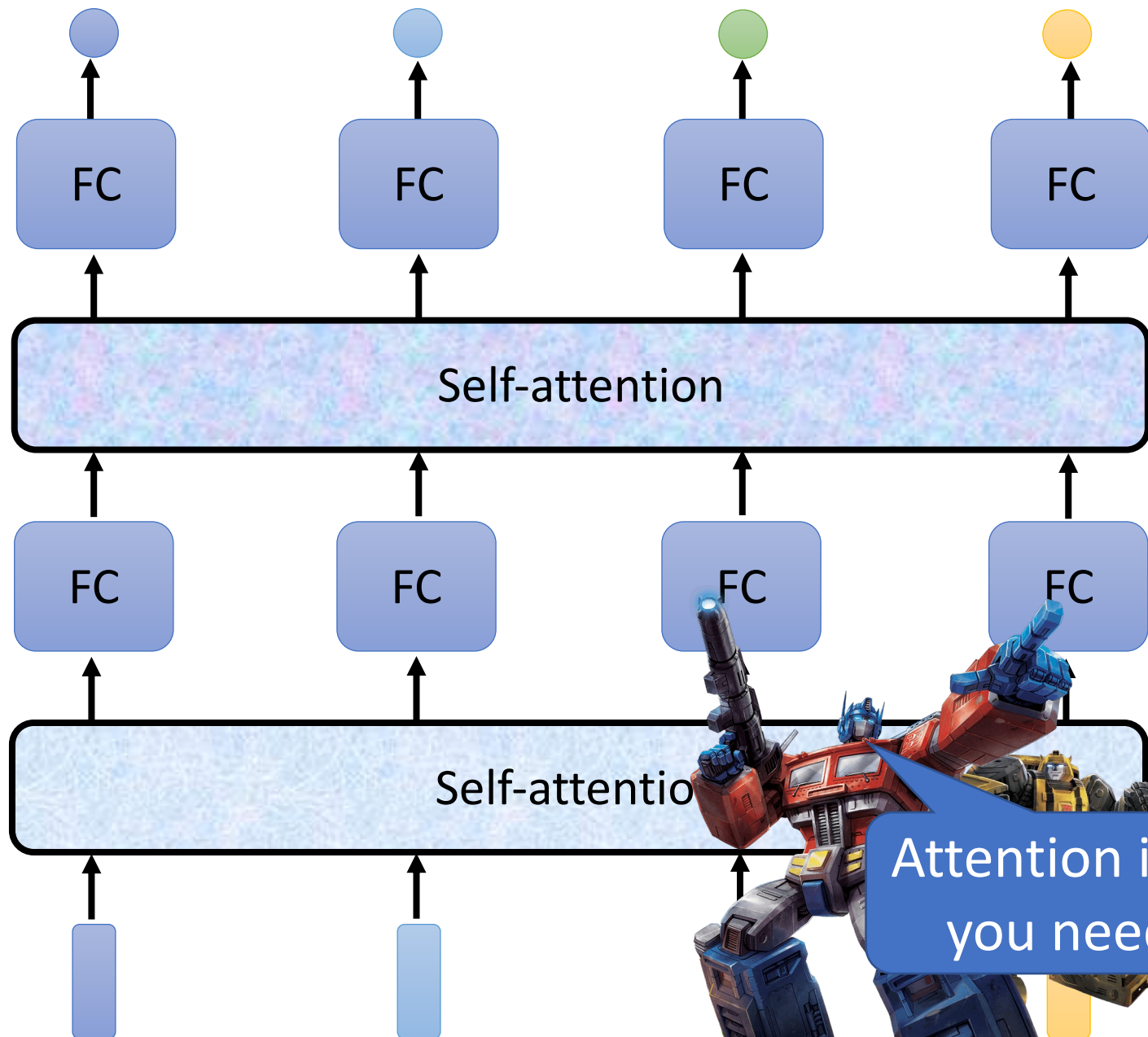
How to consider the whole sequence?

a window covers the whole sequence?

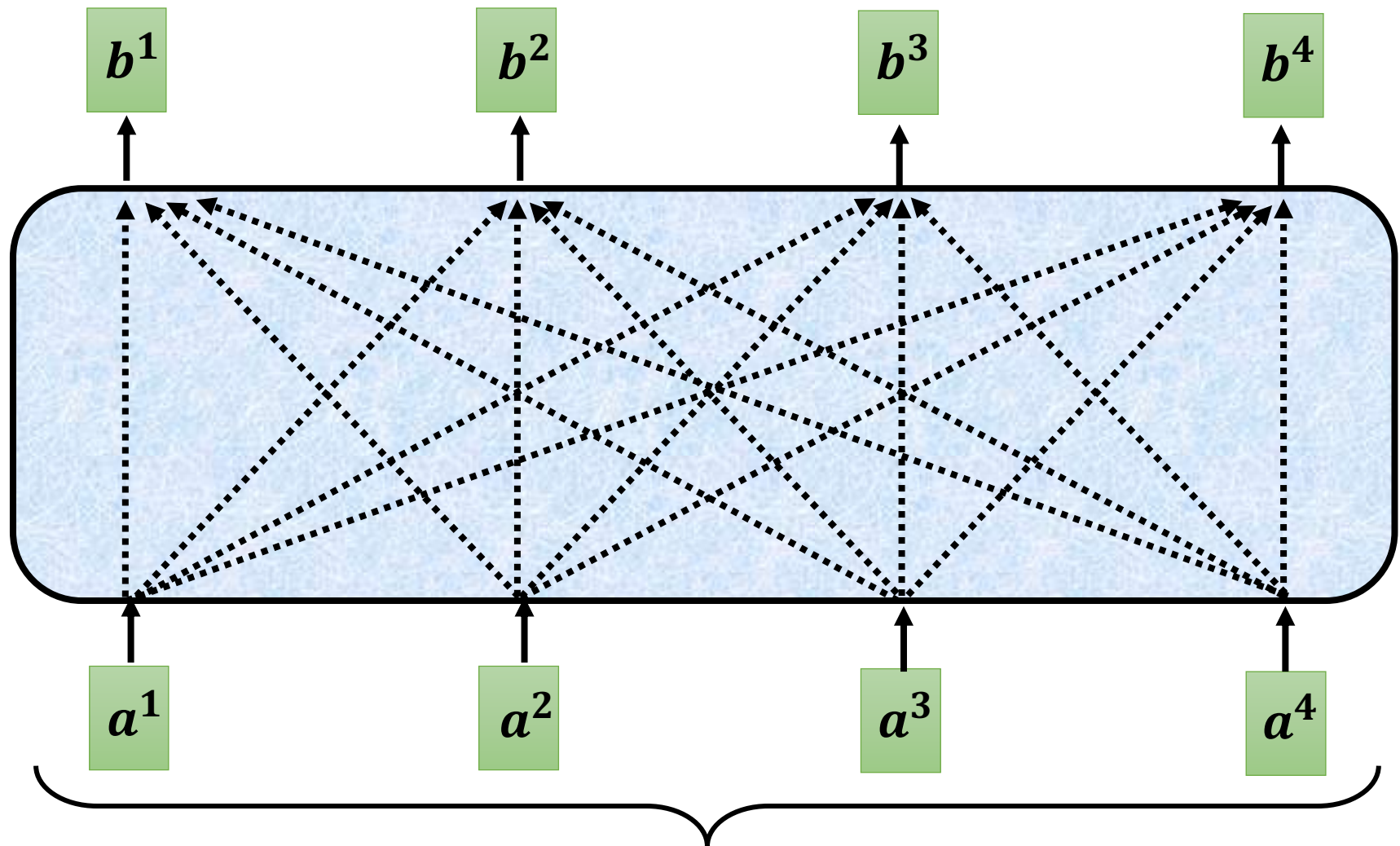


Self-attention



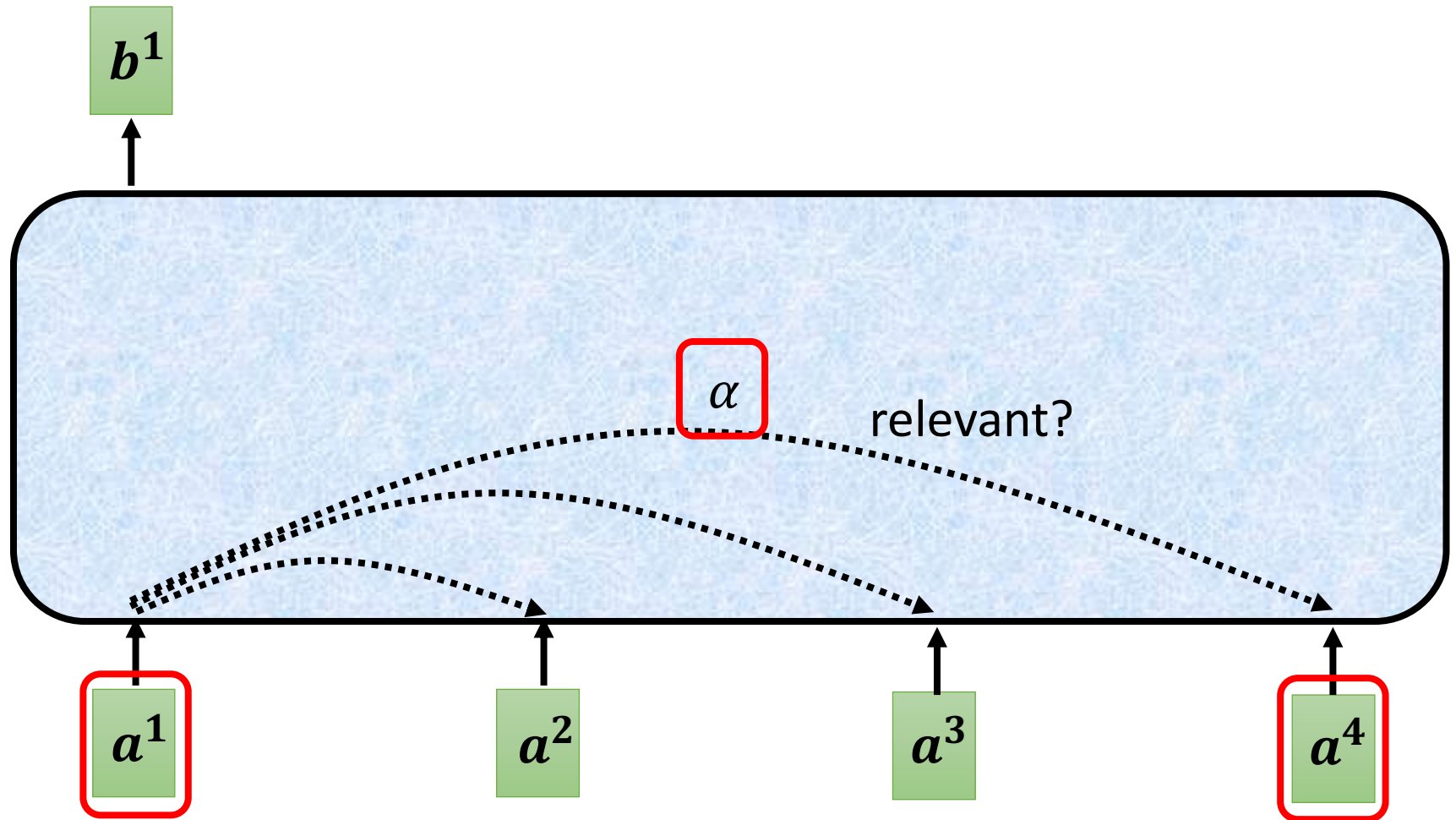


Self-attention



Can be either **input** or a **hidden layer**

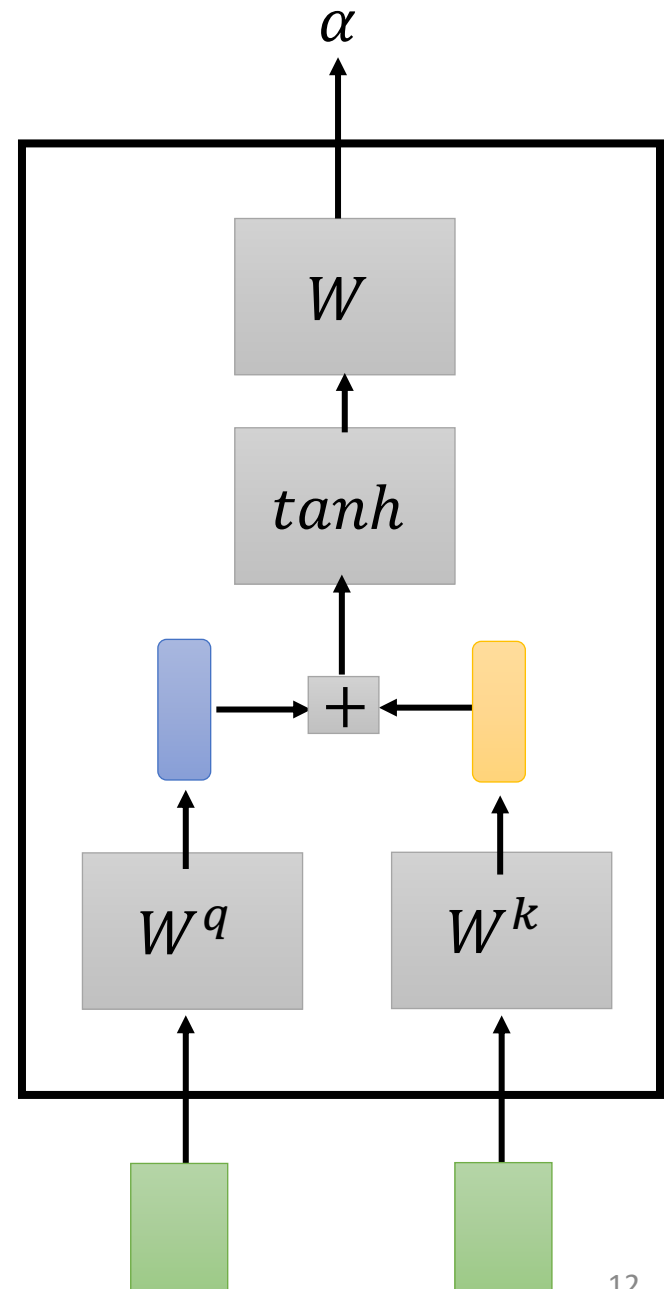
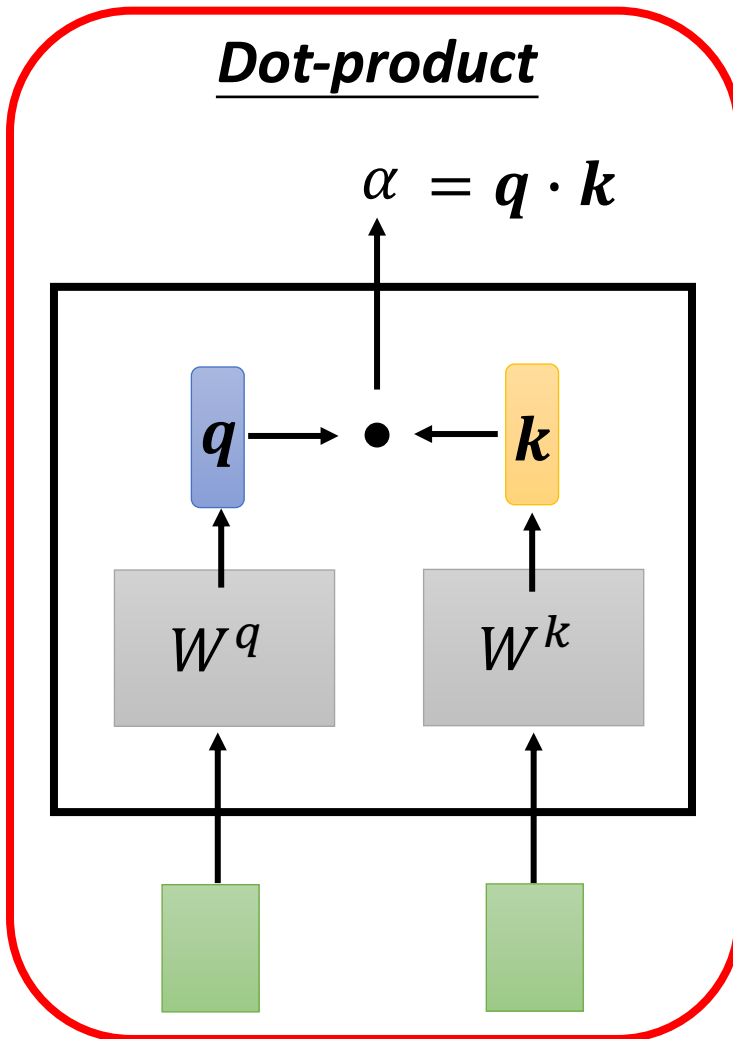
Self-attention



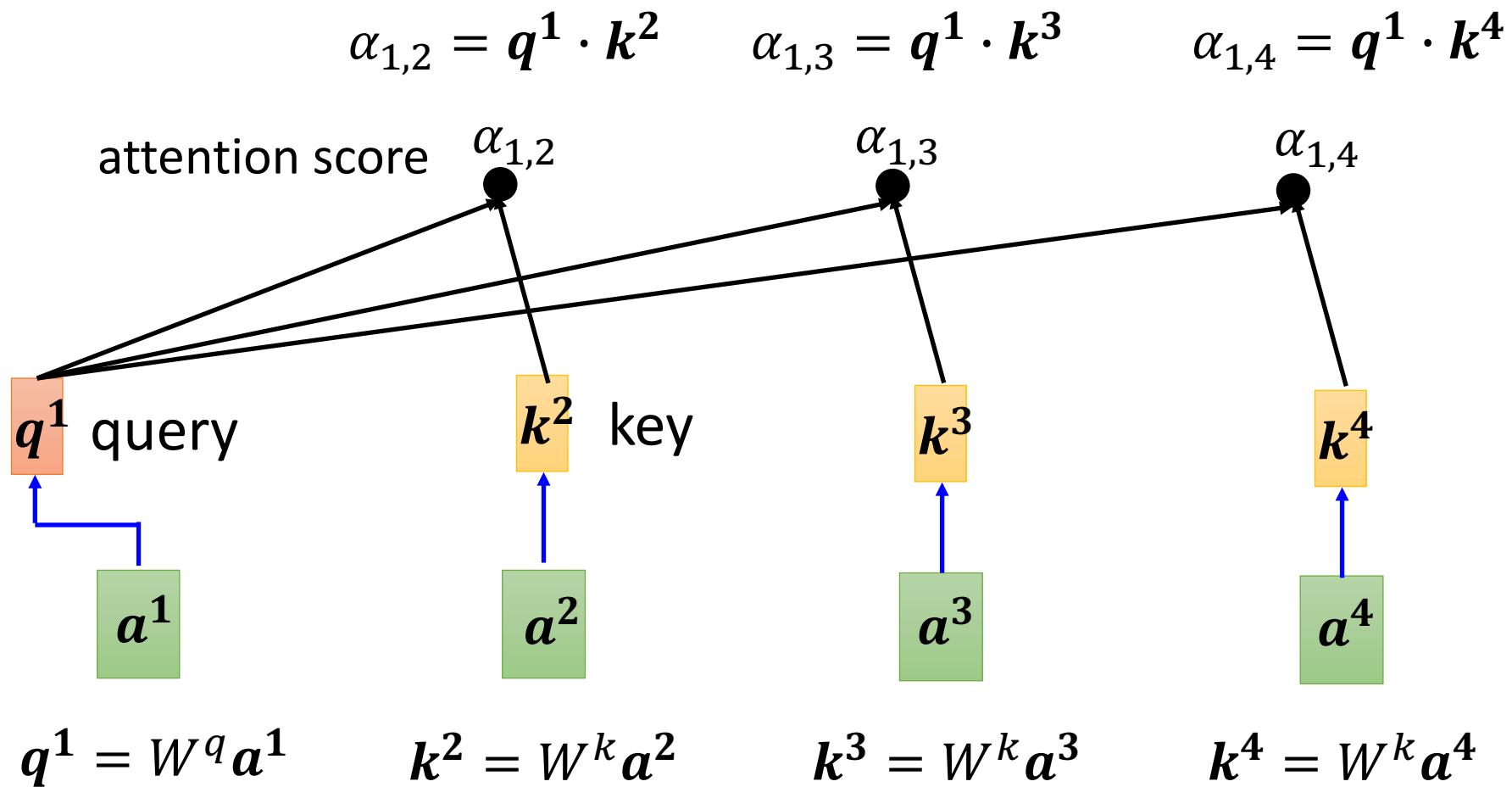
Find the relevant vectors in a sequence

Self-attention

Additive

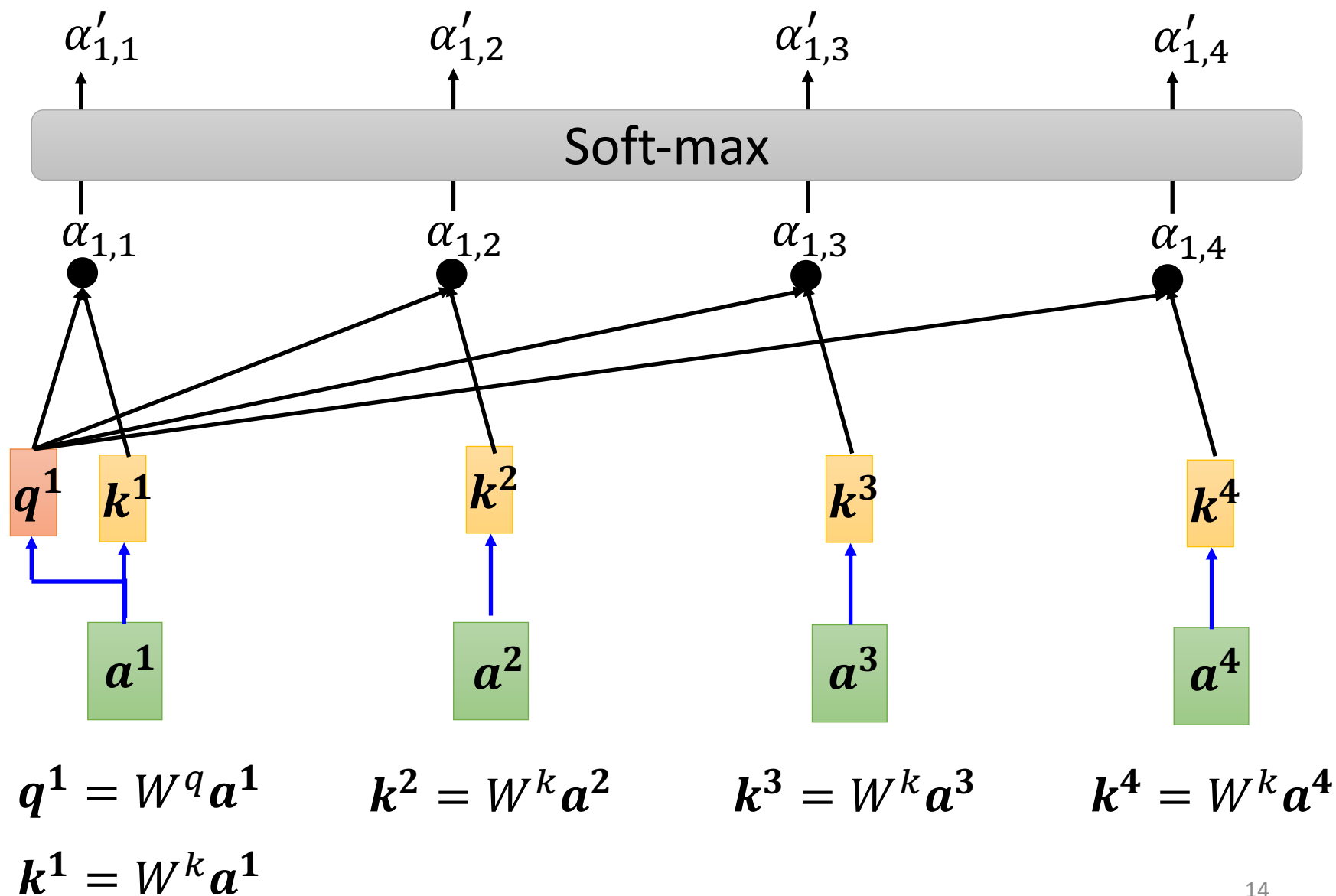


Self-attention



Self-attention

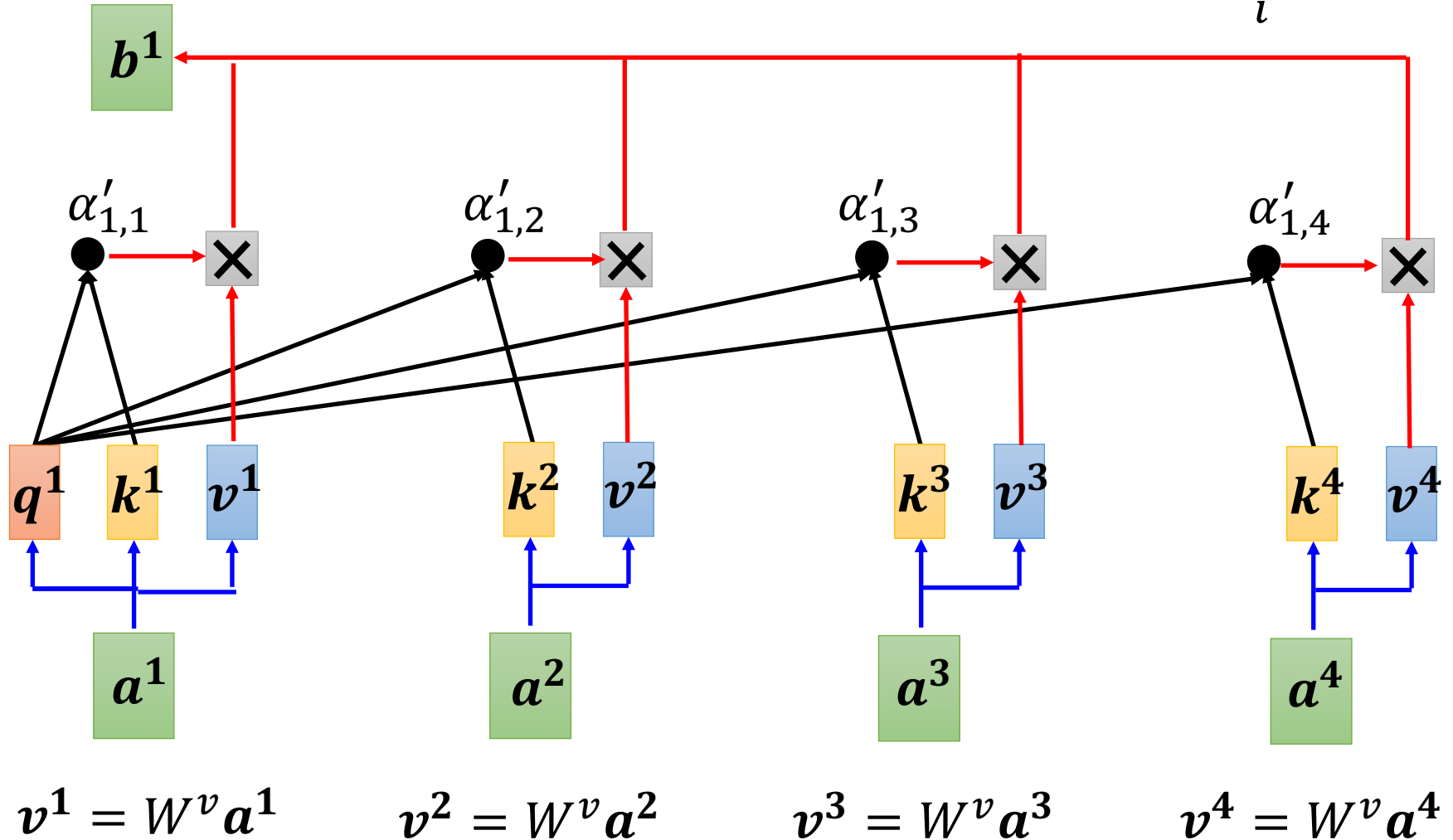
$$\alpha'_{1,i} = \exp(\alpha_{1,i}) / \sum_j \exp(\alpha_{1,j})$$



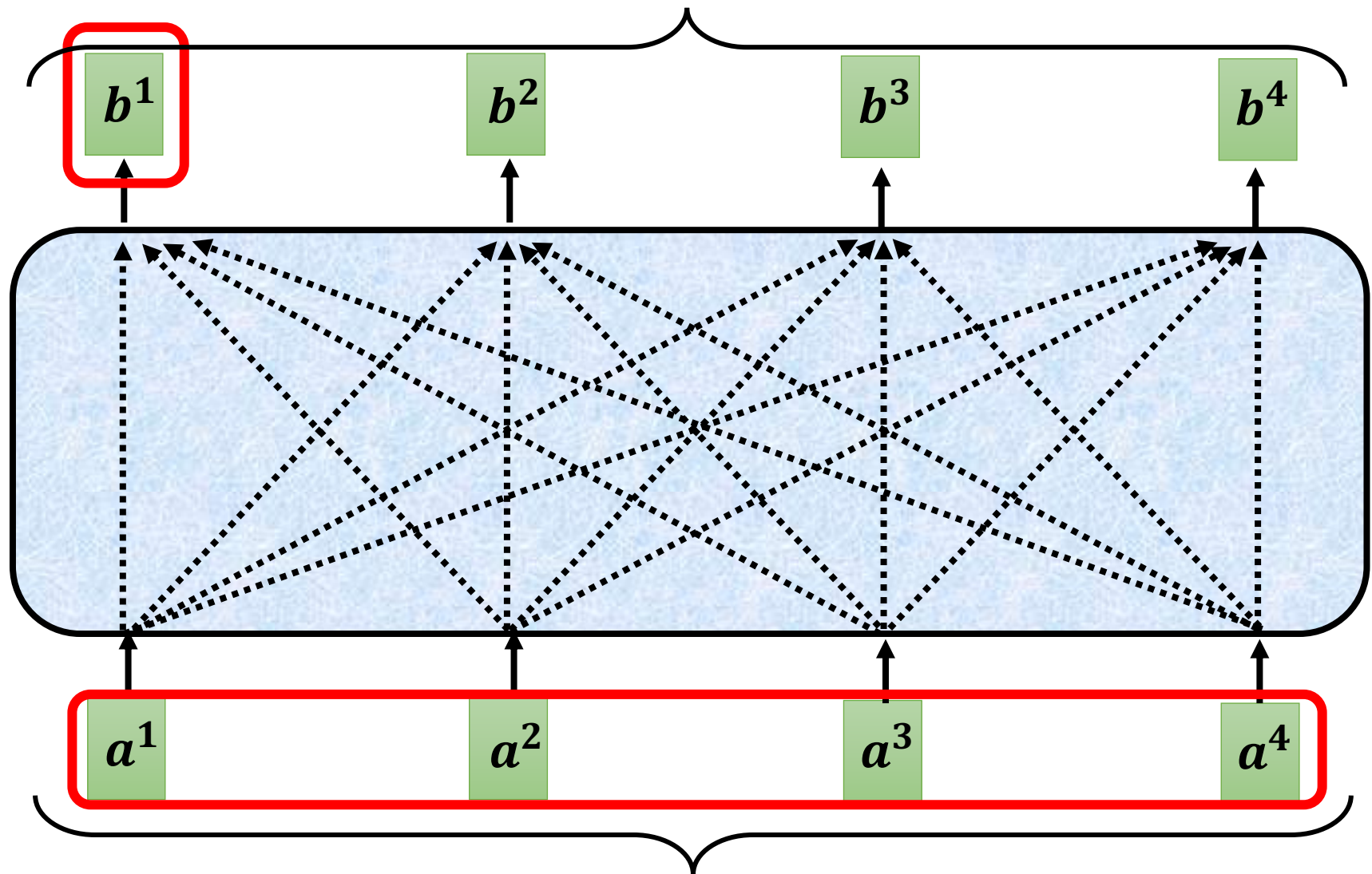
Self-attention

Extract information based on attention scores

$$b^1 = \sum_i \alpha'_{1,i} v^i$$



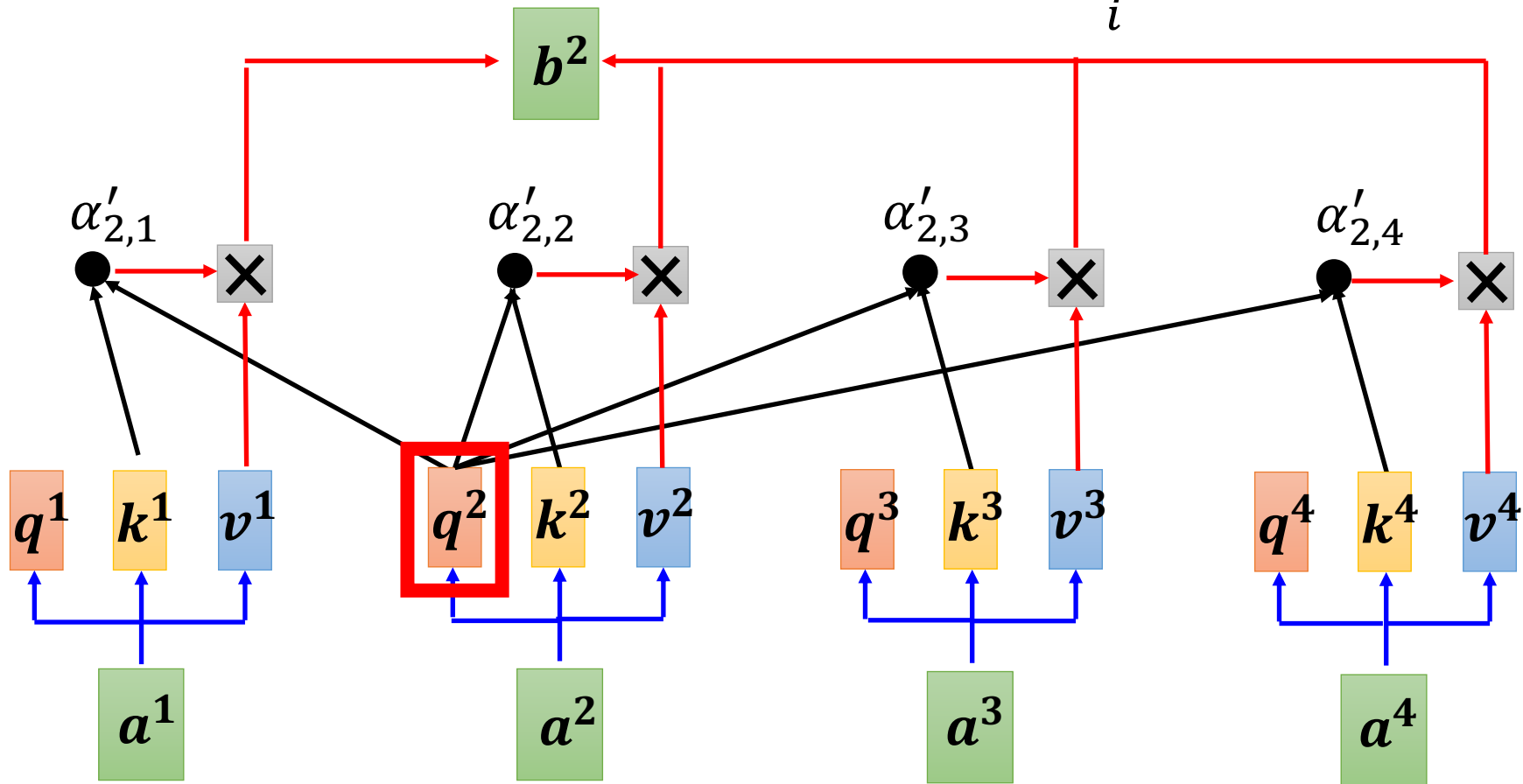
Self-attention



Can be either **input** or a **hidden layer**

Self-attention

$$b^2 = \sum_i \alpha'_{2,i} v^i$$



Self-attention

$$q^i = W^q a^i$$

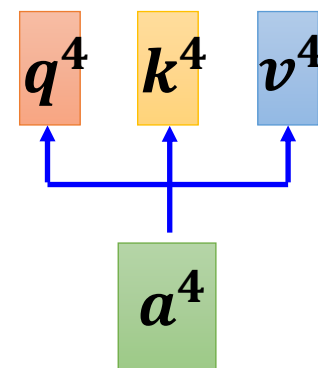
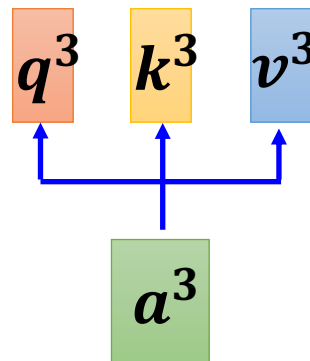
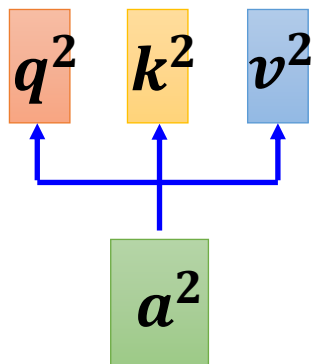
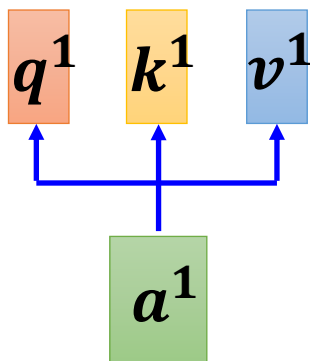
$$\begin{array}{c} q^1 q^2 q^3 q^4 \\ Q \end{array} = \begin{array}{c} W^q \\ \text{I} \end{array} \begin{array}{c} a^1 a^2 a^3 a^4 \\ \text{I} \end{array}$$

$$k^i = W^k a^i$$

$$\begin{array}{c} k^1 k^2 k^3 k^4 \\ K \end{array} = \begin{array}{c} W^k \\ \text{I} \end{array} \begin{array}{c} a^1 a^2 a^3 a^4 \\ \text{I} \end{array}$$

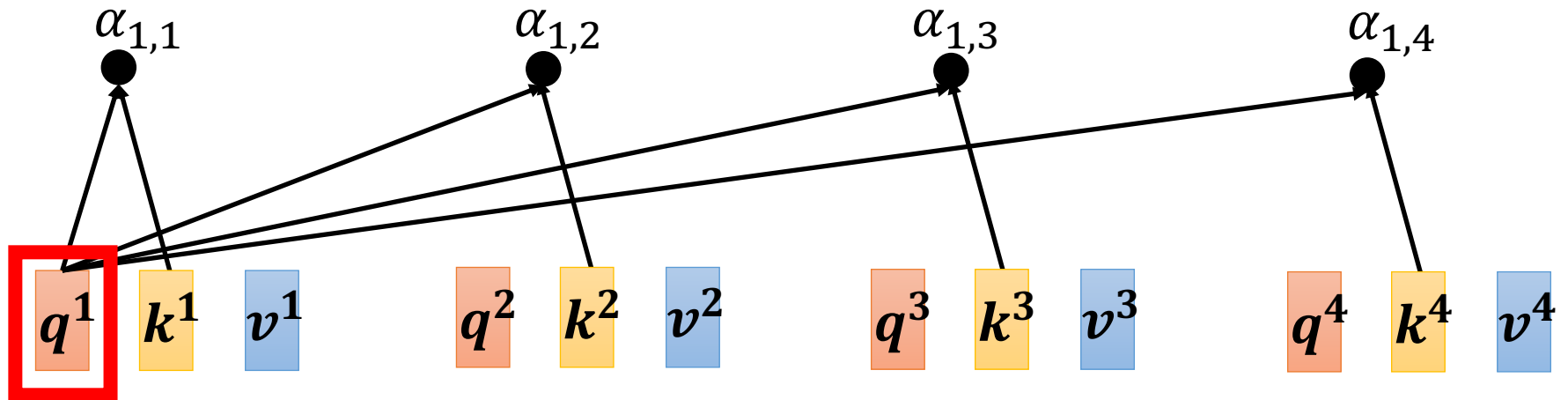
$$v^i = W^v a^i$$

$$\begin{array}{c} v^1 v^2 v^3 v^4 \\ V \end{array} = \begin{array}{c} W^v \\ \text{I} \end{array} \begin{array}{c} a^1 a^2 a^3 a^4 \\ \text{I} \end{array}$$



Self-attention

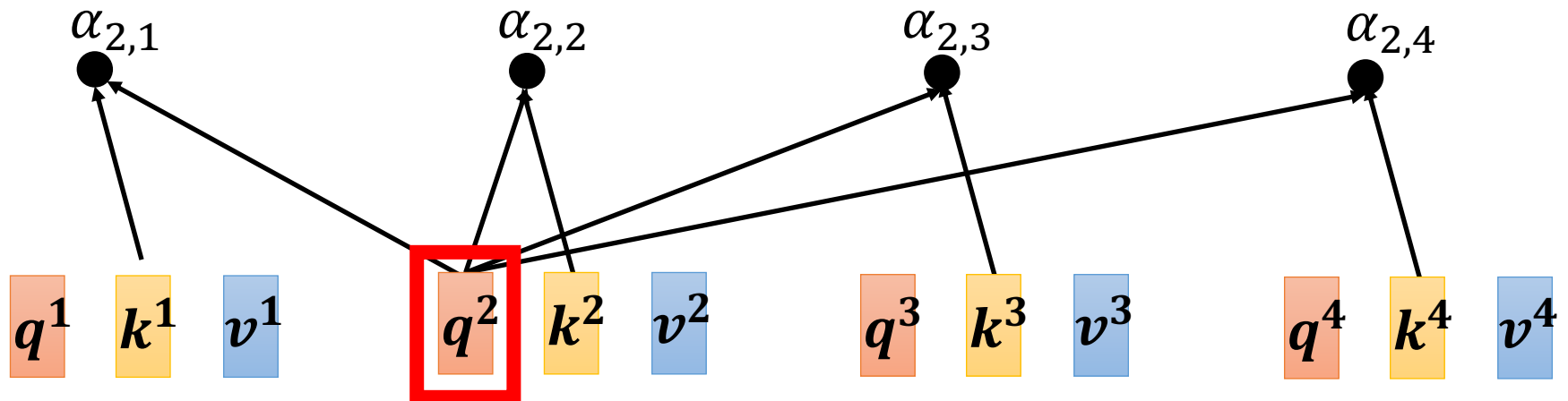
$$\begin{aligned}\alpha_{1,1} &= k^1 q^1 & \alpha_{1,2} &= k^2 q^1 \\ \alpha_{1,3} &= k^3 q^1 & \alpha_{1,4} &= k^4 q^1\end{aligned}$$
$$\begin{matrix} \alpha_{1,1} \\ \alpha_{1,2} \\ \alpha_{1,3} \\ \alpha_{1,4} \end{matrix} = \begin{matrix} k^1 \\ k^2 \\ k^3 \\ k^4 \end{matrix} q^1$$



Self-attention

$$\begin{aligned}\alpha_{1,1} &= k^1 q^1 & \alpha_{1,2} &= k^2 q^1 \\ \alpha_{1,3} &= k^3 q^1 & \alpha_{1,4} &= k^4 q^1\end{aligned}$$

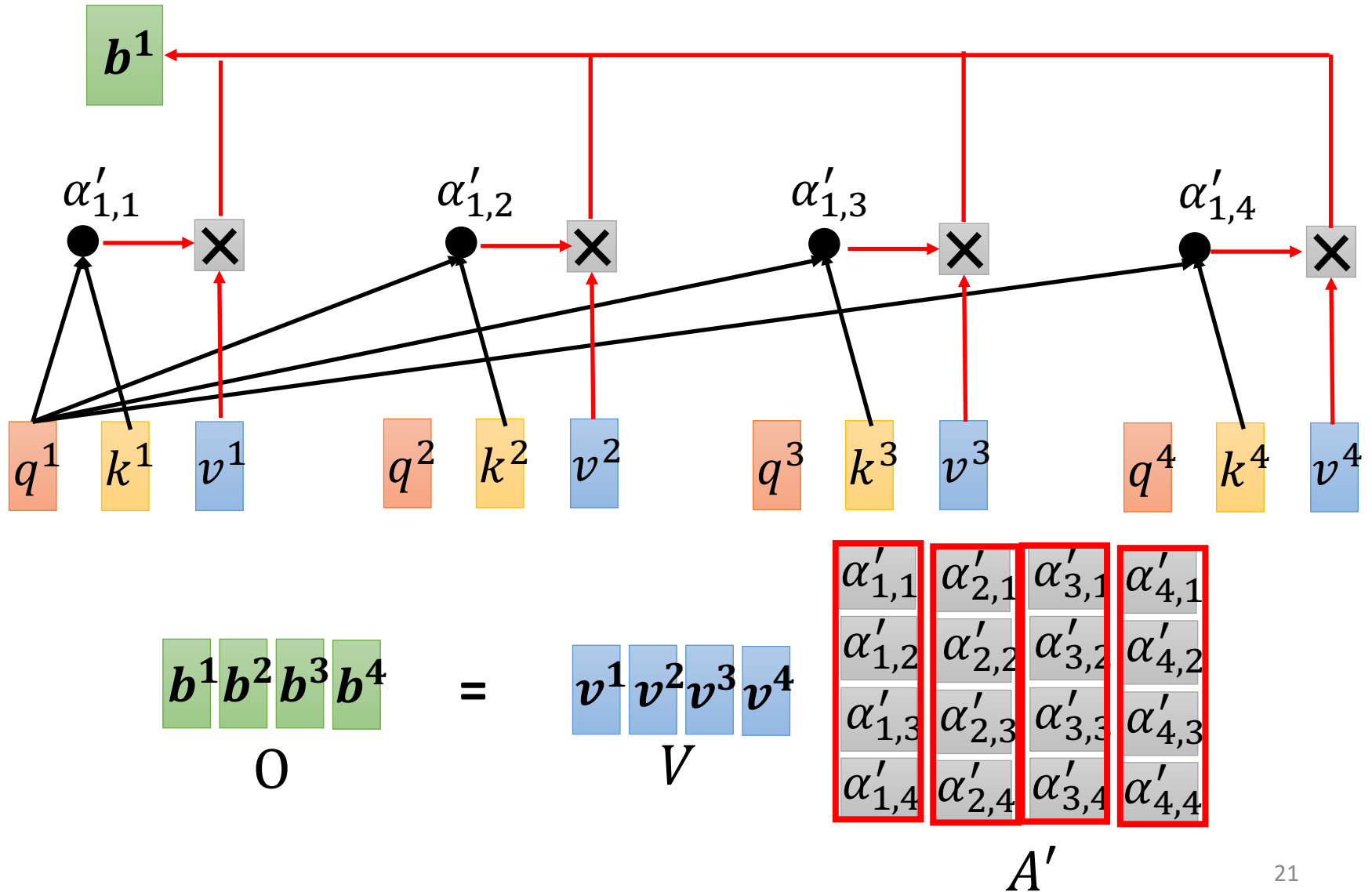
$$\begin{bmatrix} \alpha_{1,1} \\ \alpha_{1,2} \\ \alpha_{1,3} \\ \alpha_{1,4} \end{bmatrix} = \begin{bmatrix} k^1 \\ k^2 \\ k^3 \\ k^4 \end{bmatrix} q^1$$



$$\begin{bmatrix} \alpha'_{1,1} & \alpha'_{2,1} & \alpha'_{3,1} & \alpha'_{4,1} \\ \alpha'_{1,2} & \alpha'_{2,2} & \alpha'_{3,2} & \alpha'_{4,2} \\ \alpha'_{1,3} & \alpha'_{2,3} & \alpha'_{3,3} & \alpha'_{4,3} \\ \alpha'_{1,4} & \alpha'_{2,4} & \alpha'_{3,4} & \alpha'_{4,4} \end{bmatrix} \leftarrow \begin{bmatrix} \alpha_{1,1} & \alpha_{2,1} & \alpha_{3,1} & \alpha_{4,1} \\ \alpha_{1,2} & \alpha_{2,2} & \alpha_{3,2} & \alpha_{4,2} \\ \alpha_{1,3} & \alpha_{2,3} & \alpha_{3,3} & \alpha_{4,3} \\ \alpha_{1,4} & \alpha_{2,4} & \alpha_{3,4} & \alpha_{4,4} \end{bmatrix} = \begin{bmatrix} k^1 \\ k^2 \\ k^3 \\ k^4 \end{bmatrix} \begin{bmatrix} q^1 & q^2 & q^3 & q^4 \end{bmatrix}$$

$A' \qquad A \qquad K^T \qquad Q$

Self-attention



Self-attention

$$\begin{aligned} Q &= W^q I \\ K &= W^k I \\ V &= W^v I \end{aligned}$$

Parameters to be learned

$$A' \leftarrow A = K^T Q$$

Attention Matrix

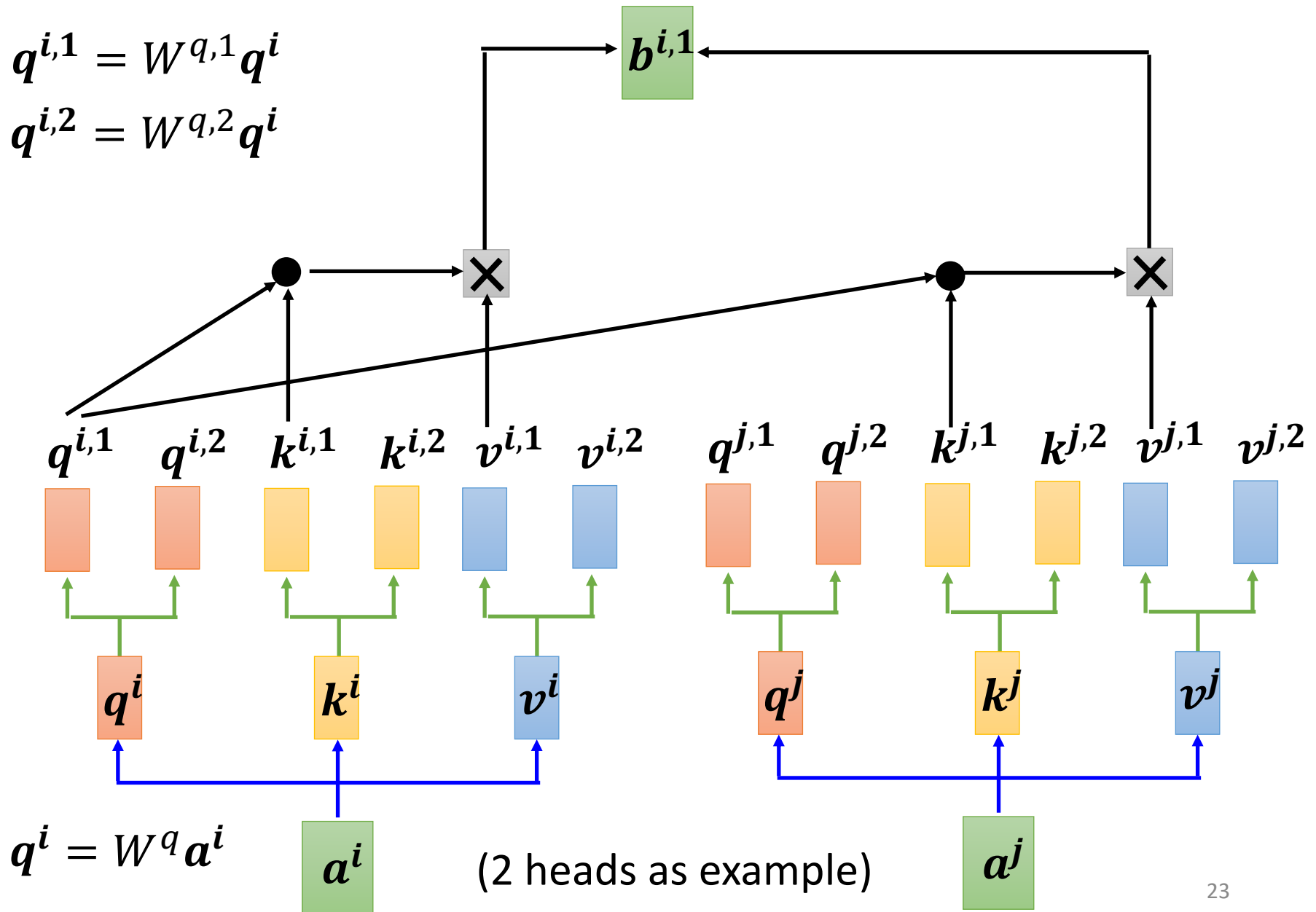
$$O = V A'$$

Multi-head Self-attention

Different types of relevance

$$q^{i,1} = W^{q,1} q^i$$

$$q^{i,2} = W^{q,2} q^i$$

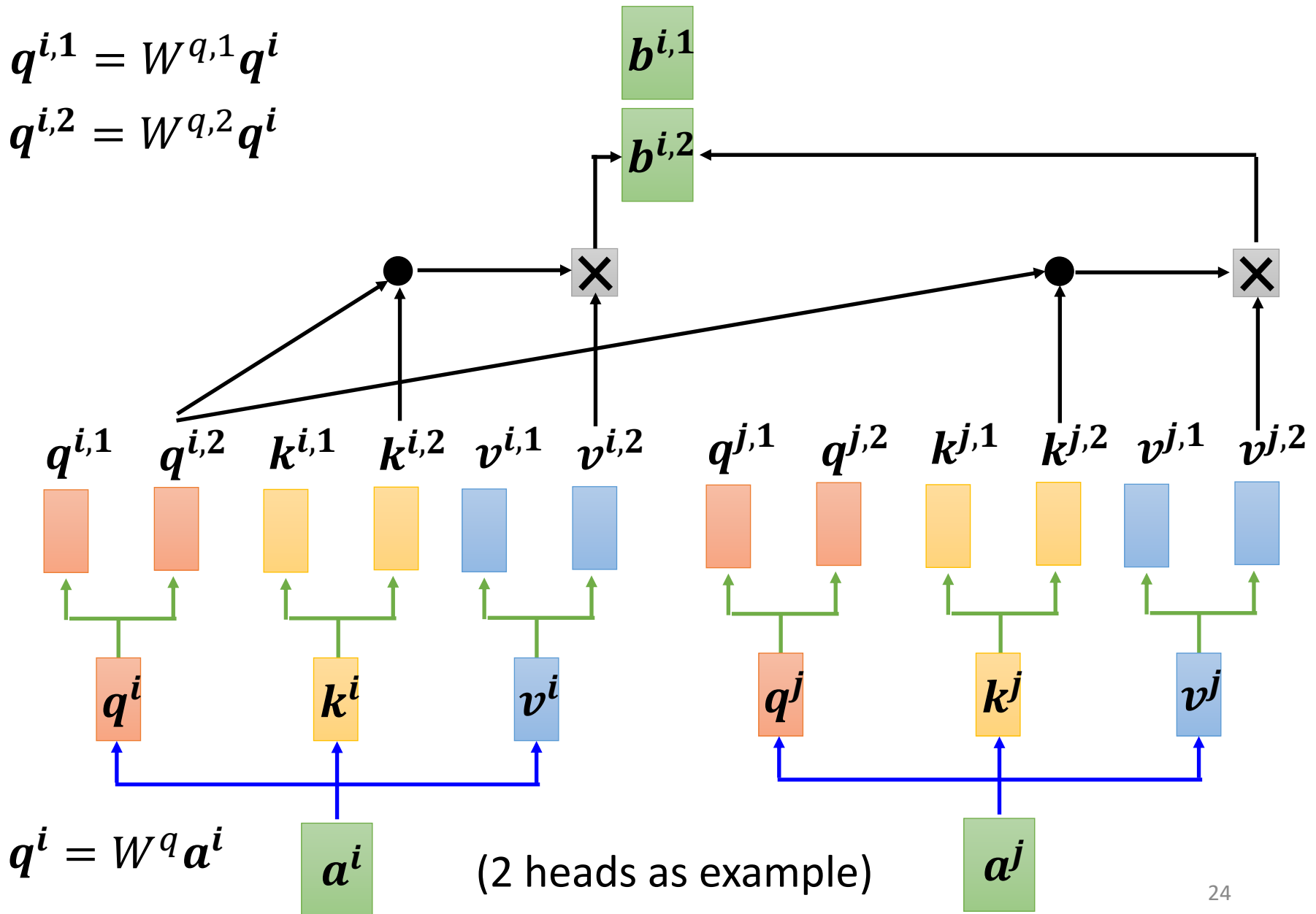


Multi-head Self-attention

Different types of relevance

$$q^{i,1} = W^{q,1} q^i$$

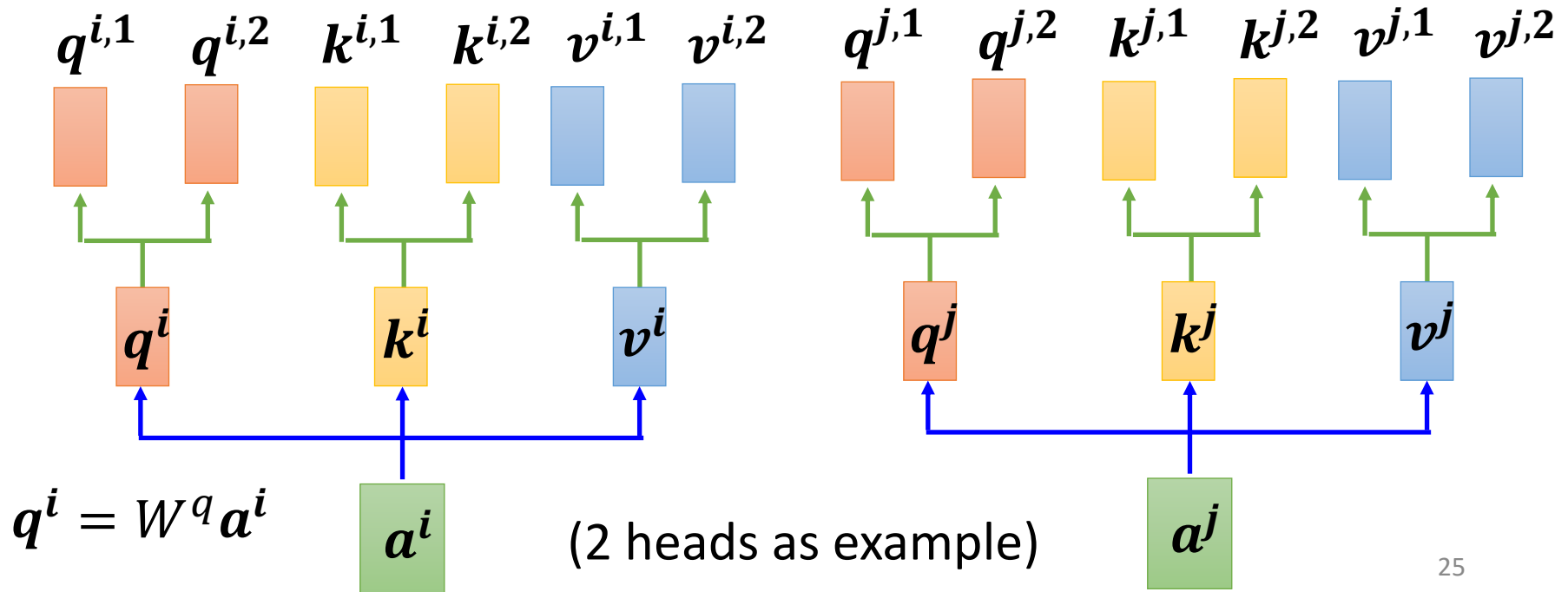
$$q^{i,2} = W^{q,2} q^i$$



Multi-head Self-attention

Different types of relevance

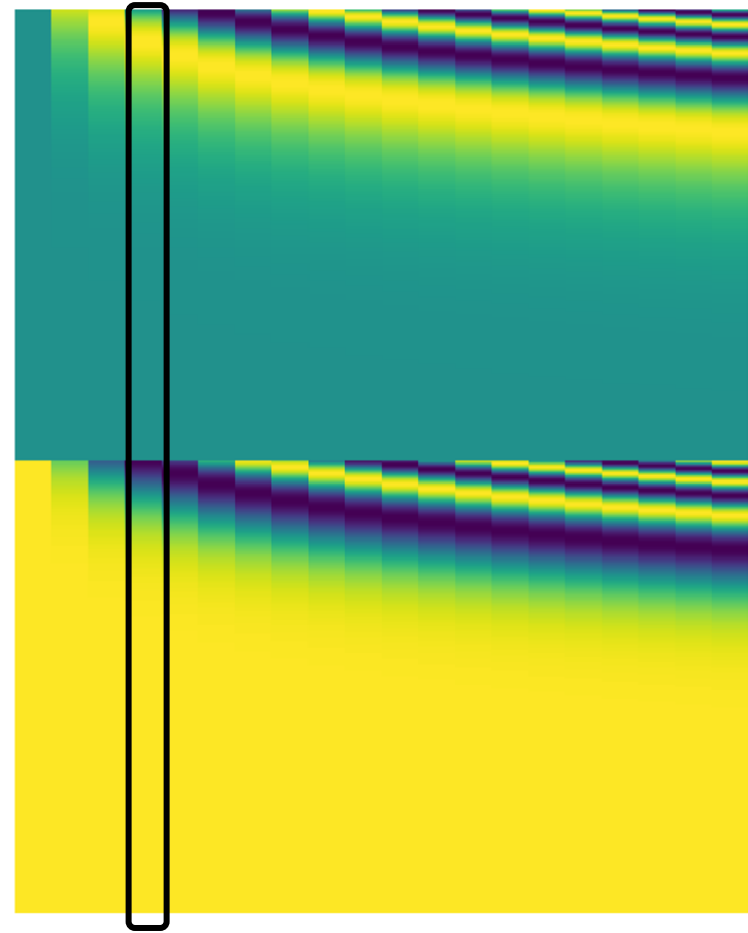
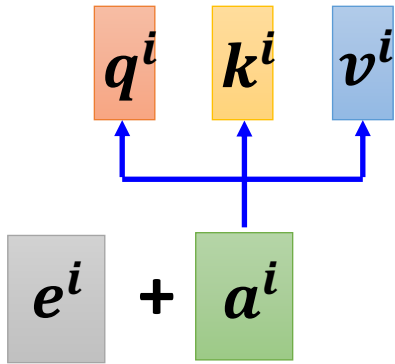
$$b^i = W^O \begin{bmatrix} b^{i,1} \\ b^{i,2} \end{bmatrix}$$



Positional Encoding

Each column represents a positional vector e^i

- No position information in self-attention.
- Each position has a unique positional vector e^i
- **hand-crafted**
- **learned from data**



-1 1

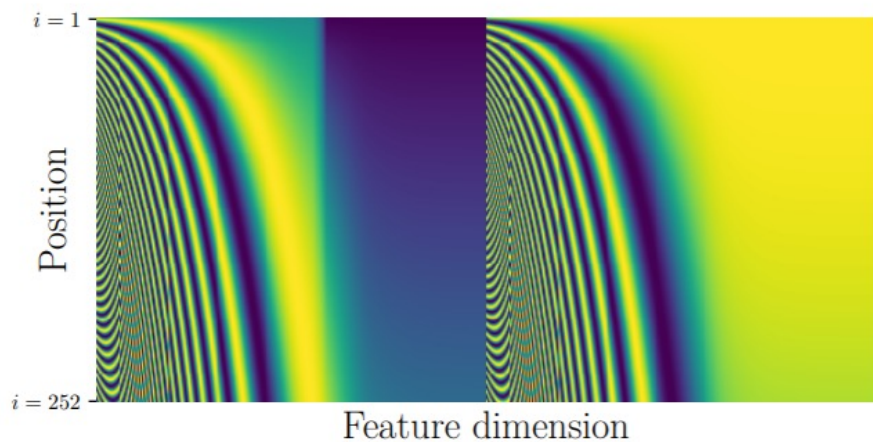
$$\text{PosEnc}(p, i) = \begin{cases} \sin\left(\frac{p}{10000^{\frac{i}{d}}}\right) & \text{if } i \text{ is even} \\ \cos\left(\frac{p}{10000^{\frac{i-1}{d}}}\right) & \text{else} \end{cases}$$

<https://arxiv.org/abs/2003.09229>

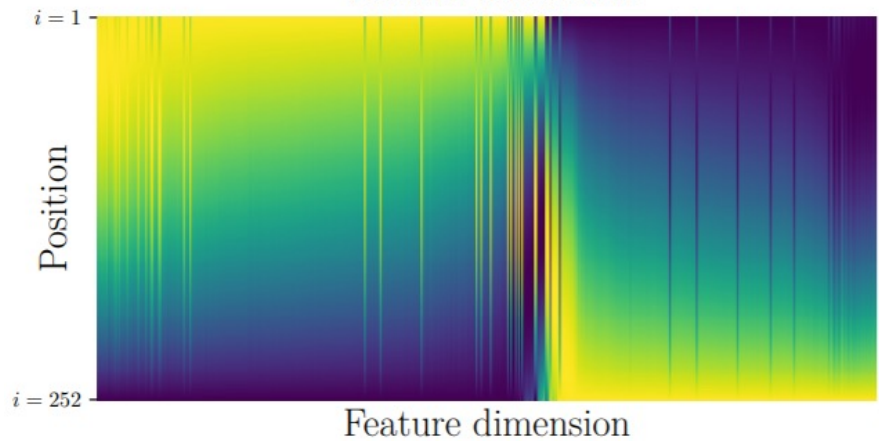
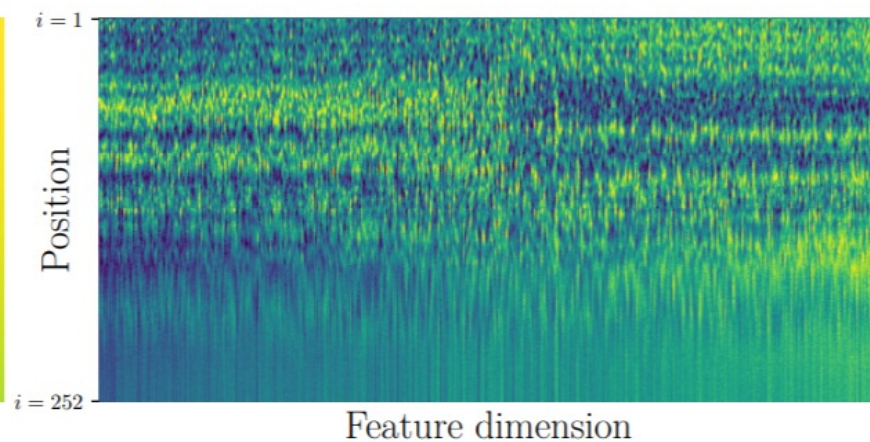
Table 1. Comparing position representation methods

Methods	Inductive	Data-Driven	Parameter Efficient
Sinusoidal (Vaswani et al., 2017)	✓	✗	✓
Embedding (Devlin et al., 2018)	✗	✓	✗
Relative (Shaw et al., 2018)	✗	✓	✓
This paper	✓	✓	✓

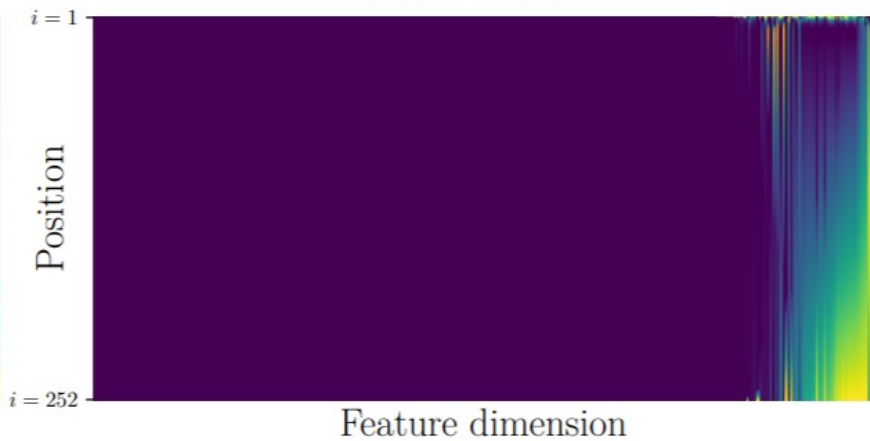
(a) Sinusoidal



(b) Position embedding



(c) FLOATER



(d) RNN

Many applications ...



Transformer

<https://arxiv.org/abs/1706.03762>



BERT

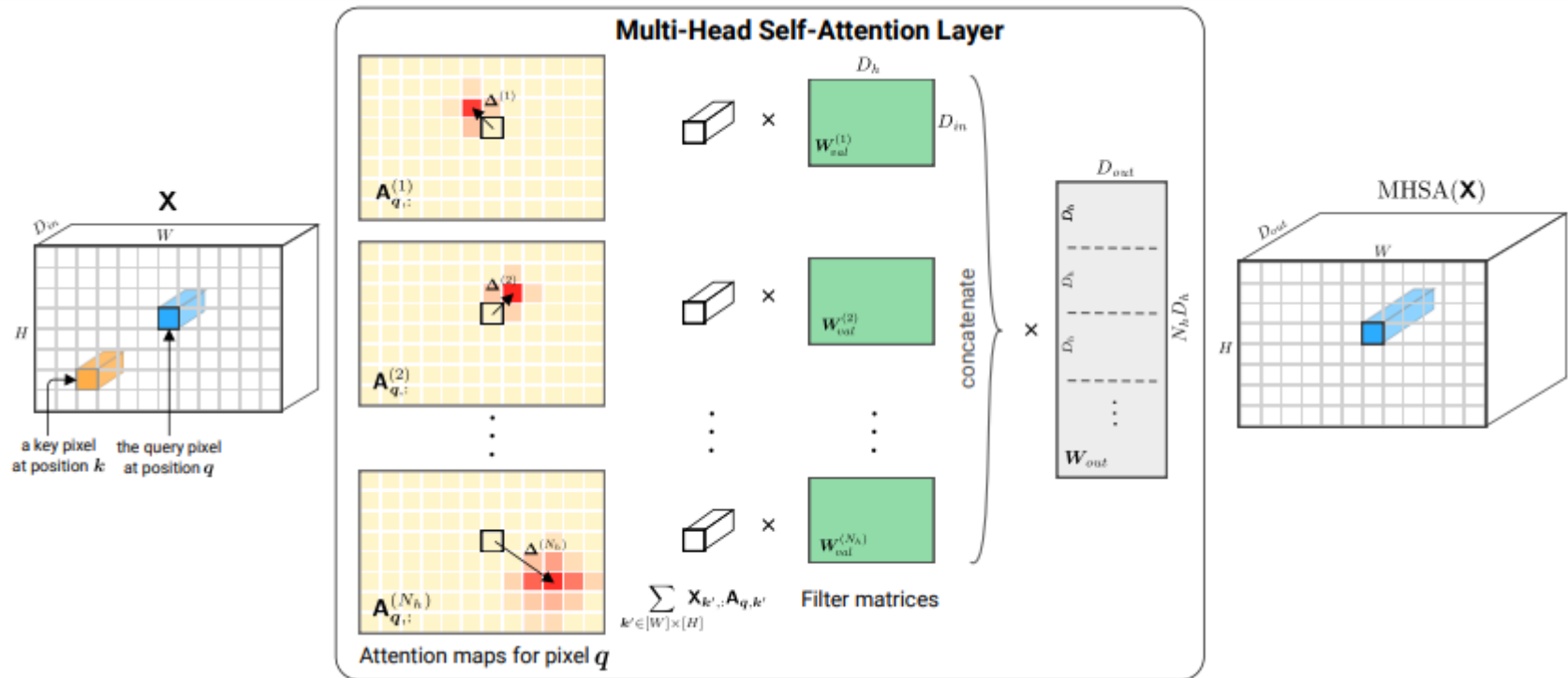
<https://arxiv.org/abs/1810.04805>

Widely used in Natural Language Processing (NLP)!

Self-attention v.s. CNN

Self-attention

CNN



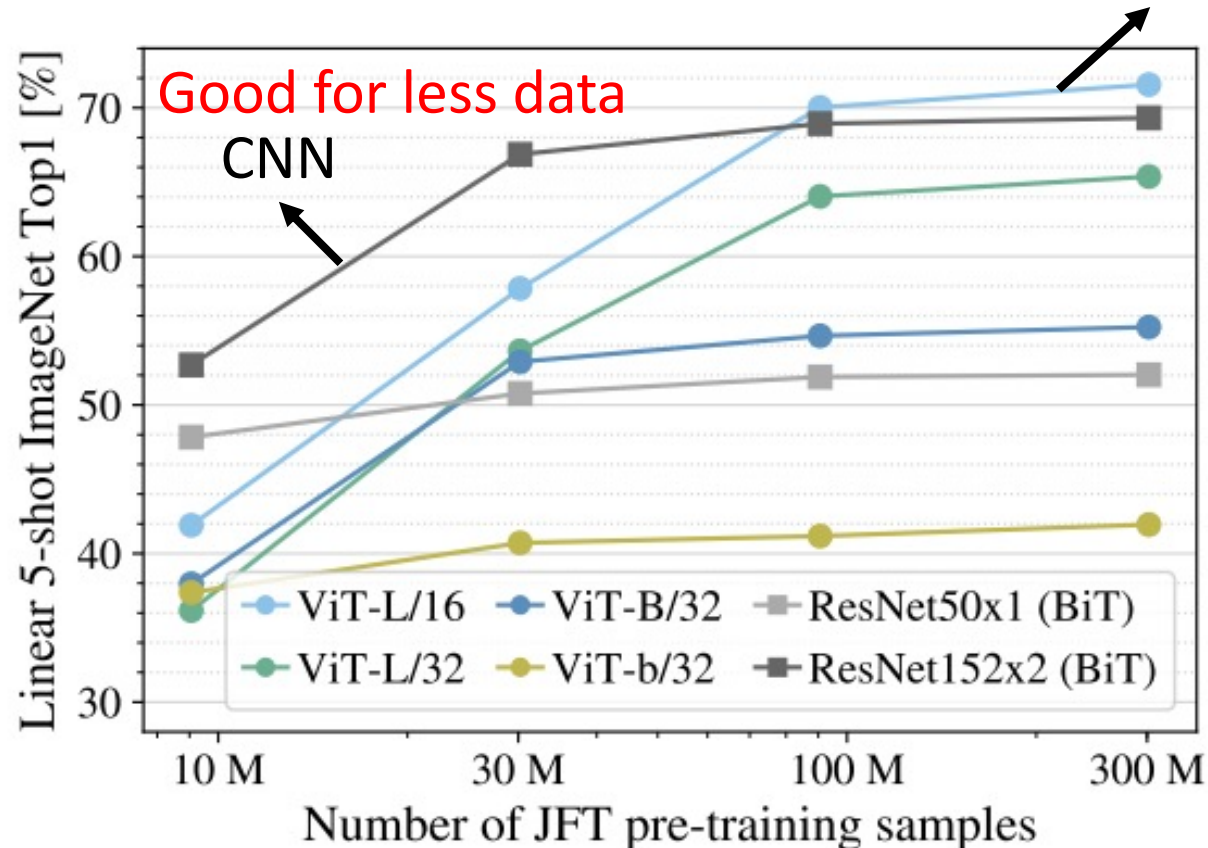
On the Relationship between Self-Attention and Convolutional Layers

<https://arxiv.org/abs/1911.03584>

Self-attention v.s. CNN

Good for more data

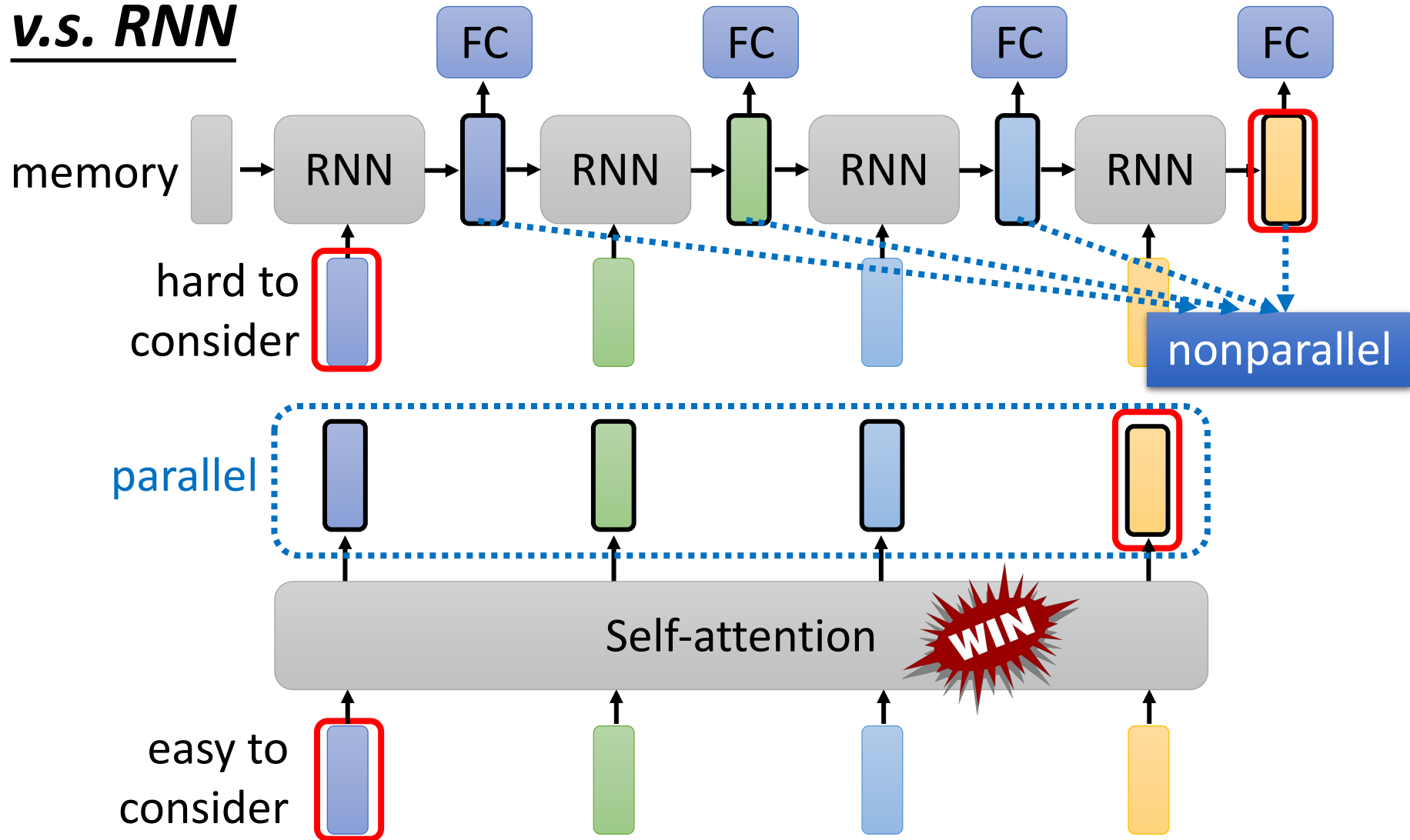
Self-attention



An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

<https://arxiv.org/pdf/2010.11929v1.pdf>

Self-attention v.s. RNN



Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention

<https://arxiv.org/abs/2006.16236>